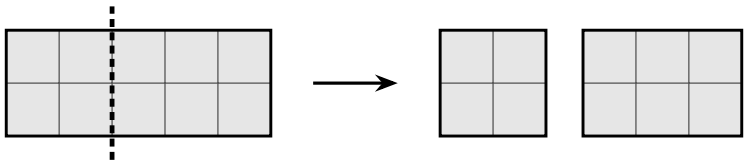
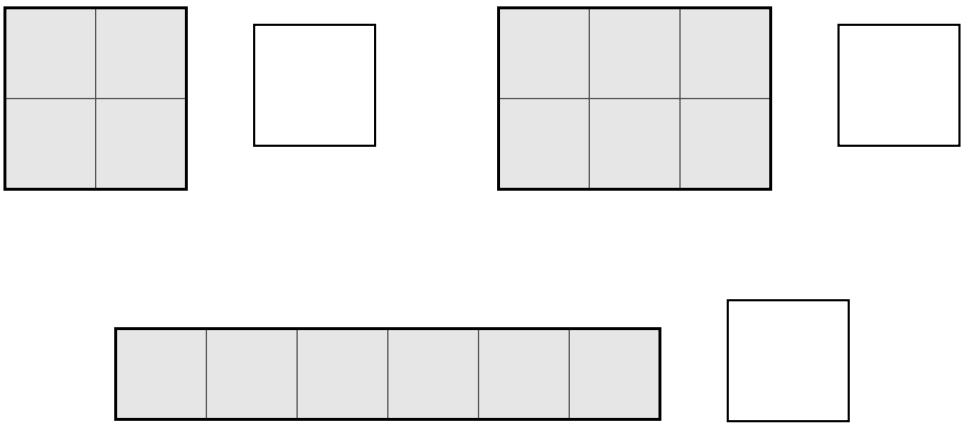
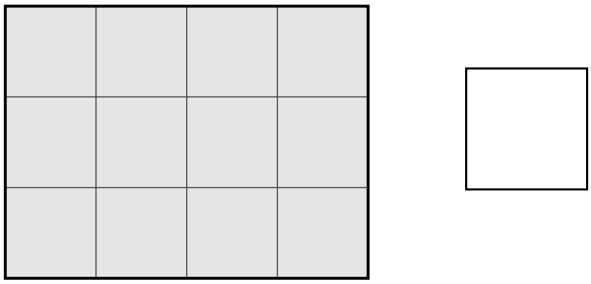
A break splits one piece into two pieces along a straight line between squares. Break one piece at a time, and never stack pieces.



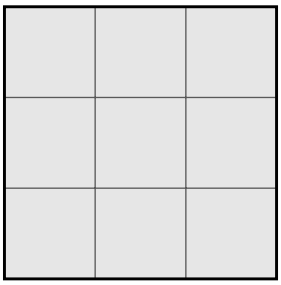
Problem 1: Break each bar into single squares. Drop a counter in the cup for each break, then write how many counters you used.



Problem 2: Break this bar into single squares using as few breaks as you can. Can anyone at your table do it with fewer breaks?

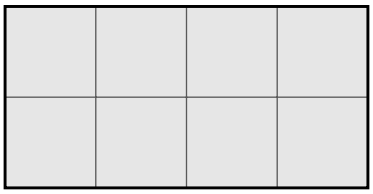


Problem 3: Play "last break wins" with a partner: take turns making one break, and whoever makes the last break wins. Play two games on each bar, with a different player going first in each game, and circle whether the 1st or 2nd player won.



Game 1 1st 2nd

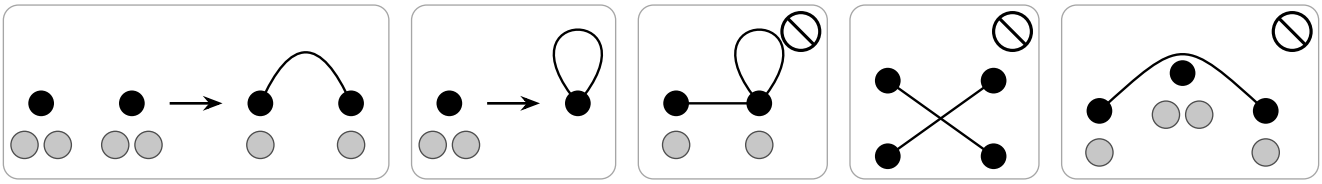
Game 2 1st 2nd



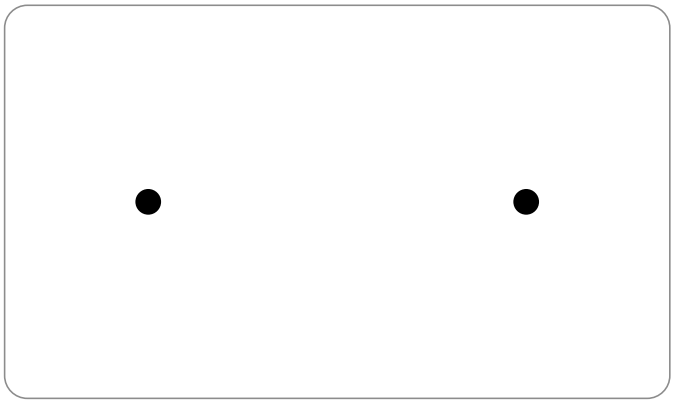
Game 1 1st 2nd

Game 2 1st 2nd

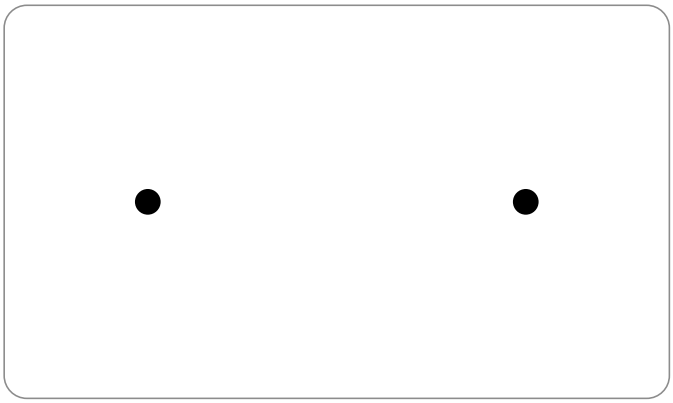
In 2-Sprigs, each dot starts with two tokens beside it. Players take turns drawing a line from a dot to another dot or back to the same dot. A line may not cross itself or another line, or touch any other dot. Each end of a line uses up one token at its dot, so a loop needs two tokens. Your partner takes away the used tokens. Whoever draws the last line wins.



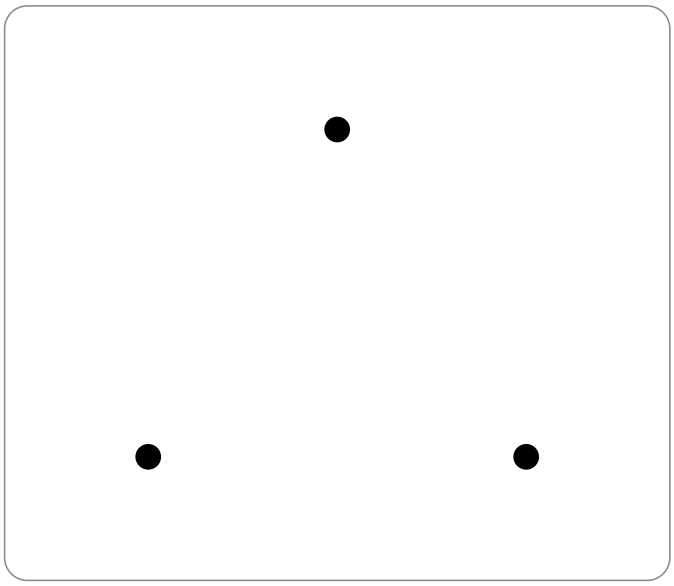
Problem 4: Play 2-Sprigs on these boards, and circle whether the 1st or 2nd player won each game.



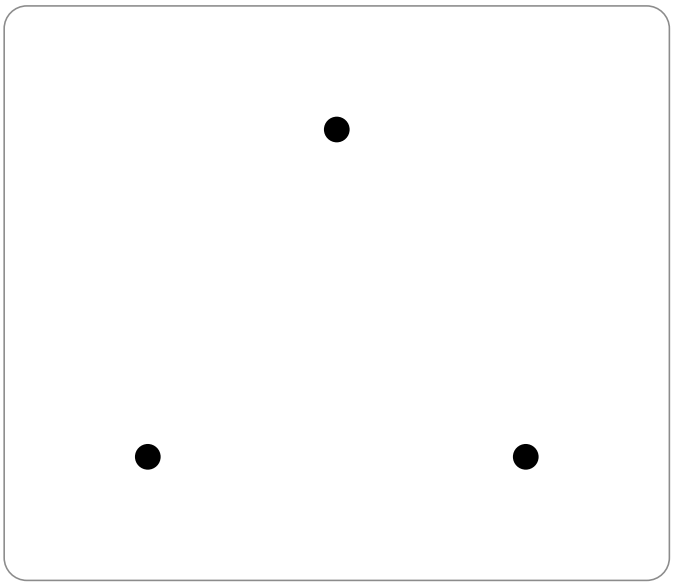
1st 2nd



1st 2nd

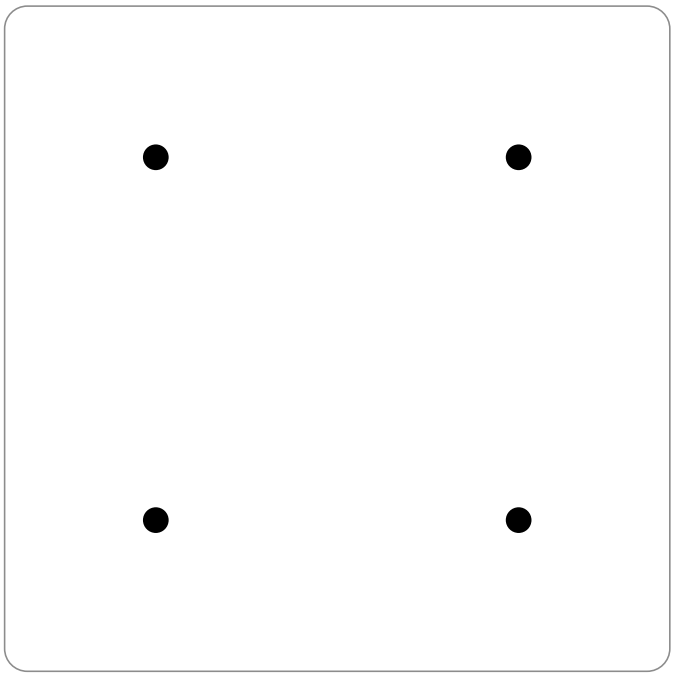


1st 2nd

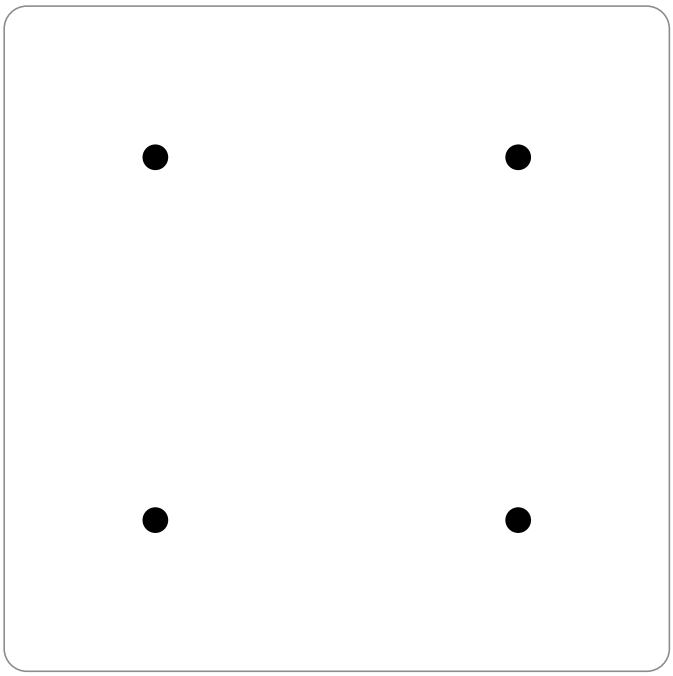


1st 2nd

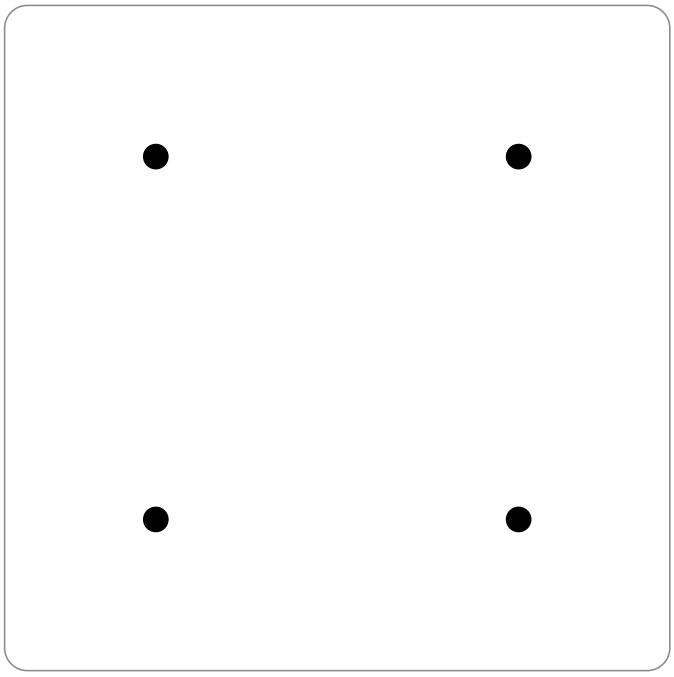
Problem 5: Play 2-Sprigs on these boards. When a game is over, count its lines and circle who won.



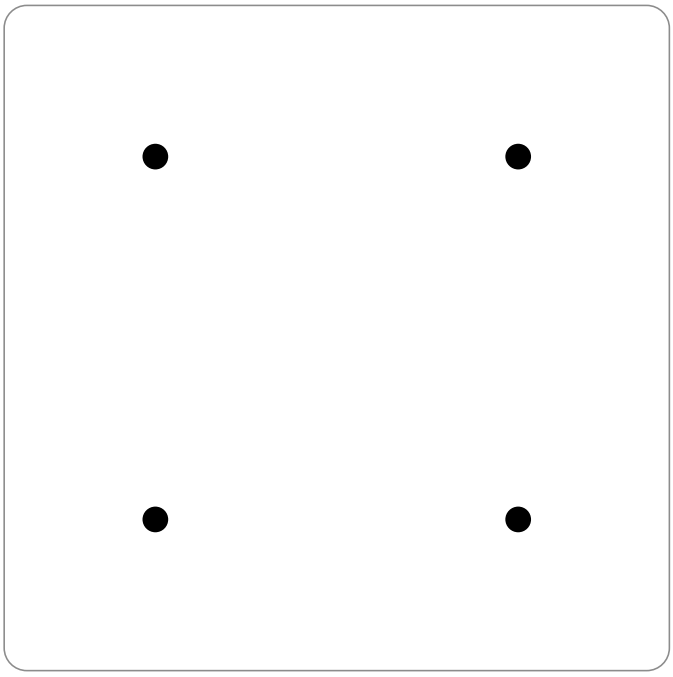
lines ☐ 1st 2nd



lines ☐ 1st 2nd

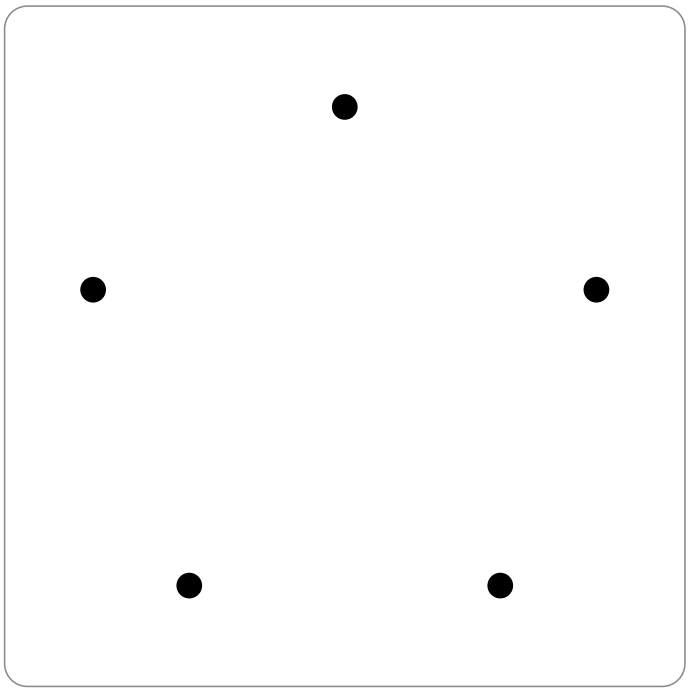


lines ☐ 1st 2nd

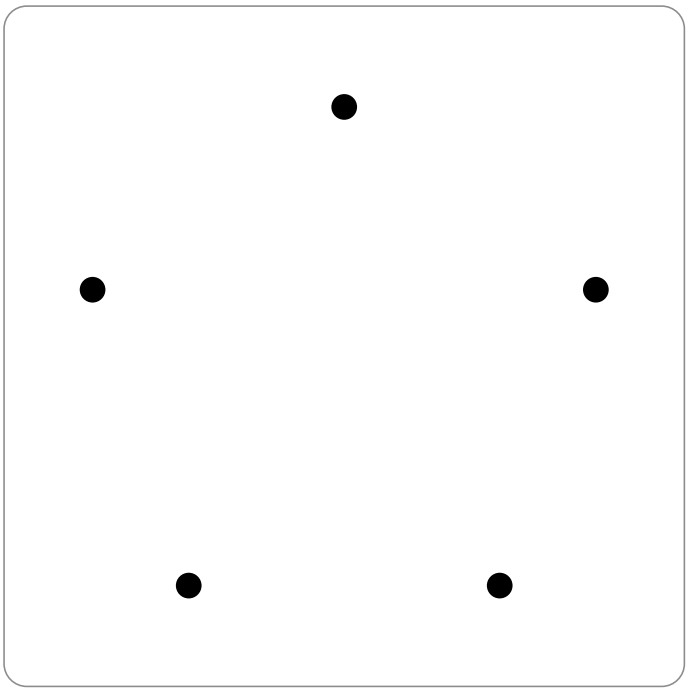


lines ☐ 1st 2nd

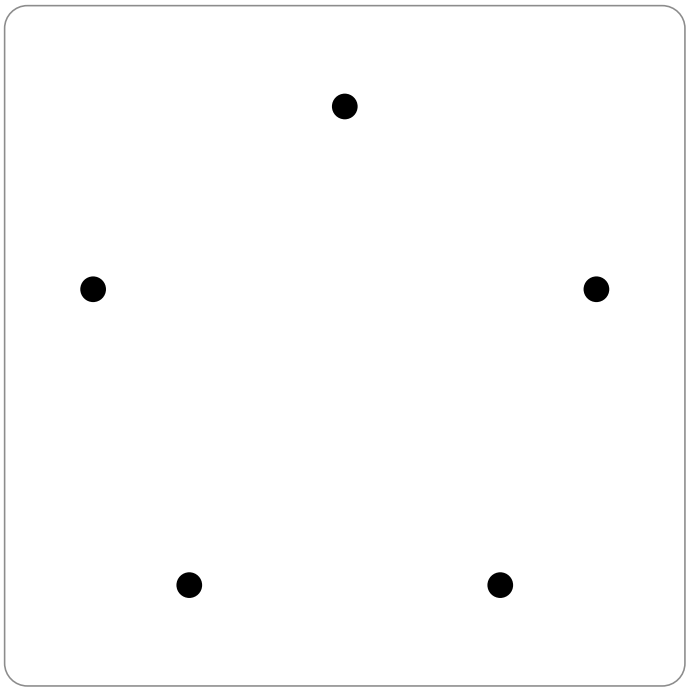
Problem 6: Can the 2nd player win 2-Sprigs on a board with 5 dots? Play to find out.



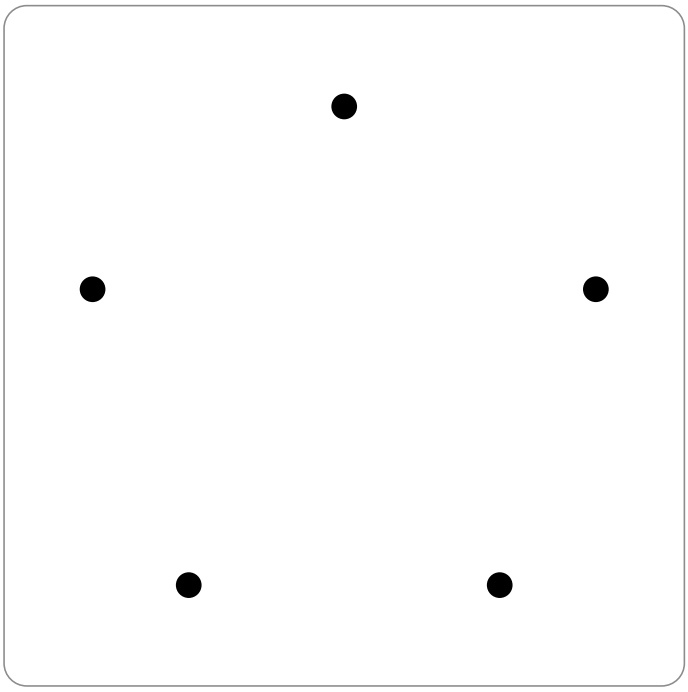
1st 2nd



1st 2nd

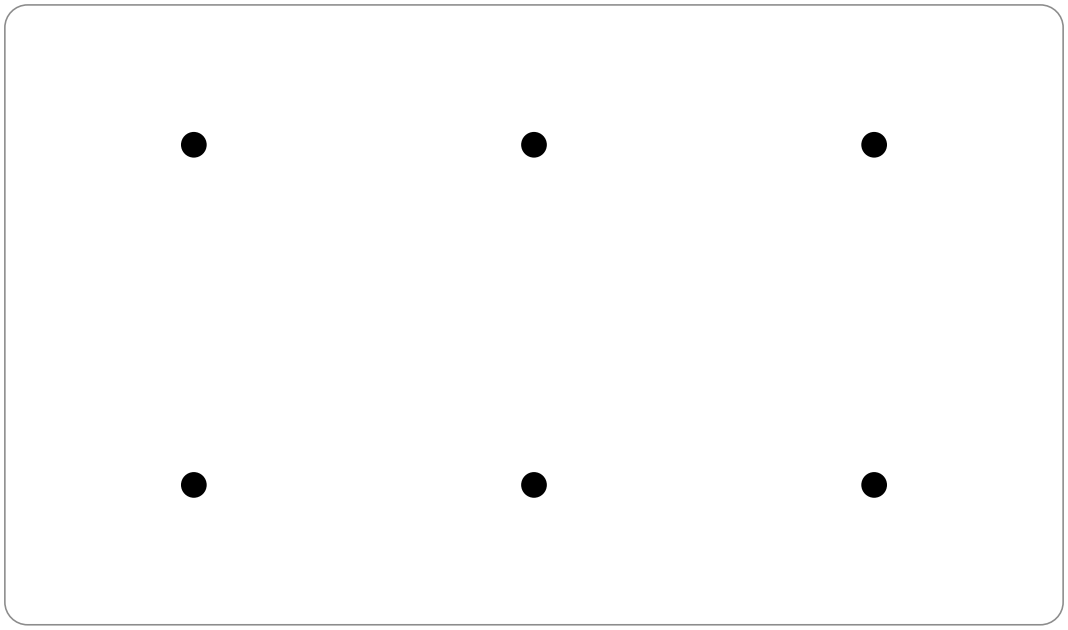


1st 2nd

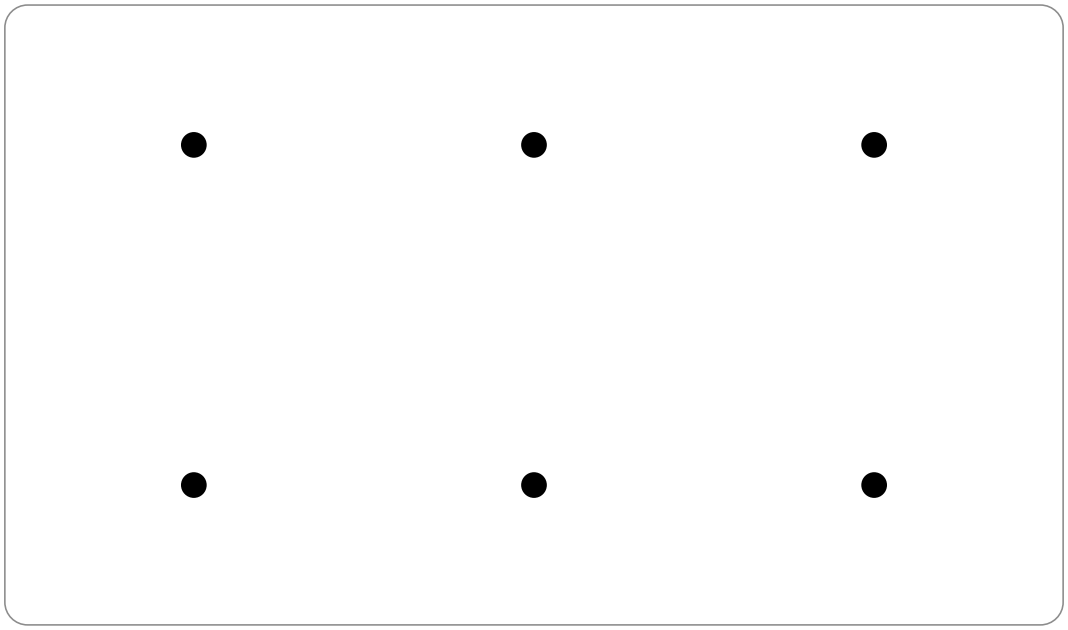


1st 2nd

Problem 7: Before you play on each board, guess whether the 1st or 2nd player will win. Then play and see.



guess 1st 2nd winner 1st 2nd



guess 1st 2nd winner 1st 2nd

Problem 8: Each group of pieces was broken from one bar. How many breaks did each group take?

